Hardware Sensor Test Documentation: Ambient Light Sensor

Module: Ambient Light Sensor Testing

Functionality: Auto-Brightness & Display Dimming

Test Case ID	Description	Pre-conditions	Test Steps	Expected Result	Status
TC_ALS_001	Verify display dimming in a completely dark environment.	Auto-Brightness toggle is enabled in Settings. Device is initially in a well-lit room.	1. Bring the device into a pitch-black room (0 lux). 2. Observe the screen brightness slider behavior.	The sensor must detect the drop in ambient light, and the OS must smoothly dim the display brightness to its minimum calibrated threshold.	**Pass**
TC_ALS_002	Verify display brightening under direct intense light (Sunlight Boost).	Auto-Brightness toggle is enabled. Device is in low ambient light.	1. Expose the top front panel of the device to direct sunlight or a strong flashlight (>10,000 lux).	The screen brightness slider must instantly scale up to 100%, activating High Brightness Mode (HBM) for outdoor legibility.	**Pass**
TC_ALS_003	Verify step-less dynamic brightness transition (Gradual Dimming).	Auto-Brightness toggle is enabled.	1. Move the device dynamically back and forth between a dimly lit corner and a bright area.	The brightness transition must be smooth, gradual, and step-less to avoid visible screen flickering or sudden harsh jumps.	**Pass**

Functionality: Native Features & App Customizations

Test Case ID	Description	Pre-conditions	Test Steps	Expected Result	Status
TC_ALS_004	Verify "Nature Tone Display" (Color Temperature Adjustment).	Nature Tone Display / Eye Comfort feature is enabled in Display Settings.	1. Move from a warm, yellowish incandescent light environment into a cool, blue-ish daylight environment.	The sensor must analyze the color spectrum of the ambient light and shift the display color temperature to match the environment.	**Pass**
TC_ALS_005	Verify camera viewfinder optimization based on ambient light.	Native Camera application is opened.	1. Point the camera viewfinder at a high-contrast backlit subject. 2. Step into a dark shadow.	The ambient light sensor data must assist the ISP framework to dynamically adjust viewfinder ISO/exposure previews before taking a snapshot.	**Pass**

Functionality: Hardware Calibration & Power States

Test Case ID	Description	Pre-conditions	Test Steps	Expected Result	Status
TC_ALS_006	Verify real-time raw Lux value variance in Engineering Mode.	Factory Diagnostic Menu (*#899# or equivalent) is launched.	1. Uncover the sensor to read room light. 2. Cover the sensor partially with a finger. 3. Note the changing numbers.	The raw value displayed in the UI must instantly change in real-time unit measurements of Lux, corresponding to actual light blocking.	**Pass**
TC_ALS_007	Verify sensor power suspension during screen off state.	Device is locked and display is completely dark.	1. Connect the device via USB and trace system logs using an ADB polling command.	The sensor subsystem must transition into a low-power sleep state, preventing continuous battery drain while the screen is inactive.	**Pass**